Algorithms

FLOW APPLICATIONS

Last Week

- Max Flow
- Ford-Fulkerson

This Week

- Edmonds Karp
- More Ford Fulkerson
- Non-trivial Max flow applications

Reminder: Max Flow

- Flow function $f: E \to R$
  - Edge Condition
  - Vertex Condition
- Flow Value F
- s – t cut $(S, V \backslash S)$
  - Flow on a cut $\sum_{e \in out(S)} f(e) - \sum_{e \in in(S)} f(e)$
  - Capacity of a cut $\sum_{e \in out(S)} c(e)$
- Residual Network N_f
- The Max Flow – Min Cut Theorem

Reminder: Max Flow Algorithm

Algorithm 1 FORD-FULKERSON(G, c)

1: Initialize flow on all edges to 0
2: **while** there is a path p from s to t in R **do**
3: $\quad \delta = \min_{e \in p} c'(e)$
4: $\quad$ Flow δ along p and update forward and backward edges accordingly
5: $\quad$ Update R along p

Time complexity:

Bound 1: $\boldsymbol{O(mf^*)}$

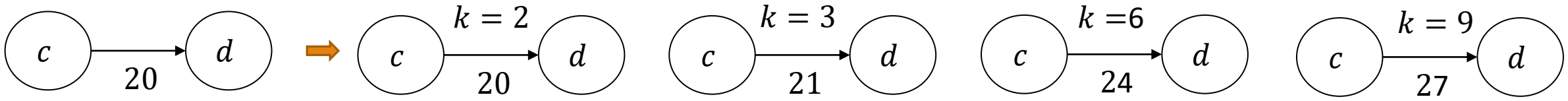
Bound 2 (EK algorithm, find p using BFS): $\boldsymbol{O(m^2n)}$

Exercise 1

Let $k > 1$ be an integer. Given a flow network with positive capacities $c: E \to \mathbb{R}^+$, every edge capacity is rounded up to the closest multiple of k.

Formally, the capacity of an edge e is determined to be

$$c'(e) = kz \text{ iff } k(z-1) < c(e) \leq kz, \text{ where } z \in \mathbb{Z}.$$



Let F denote the max-flow value in the original network.

Let F' denote the max flow value in the network after the capacity modification.

Exercise 1

Prove or disprove:

1. **F' is a multiple of k.**
2. $F \leq F'$ for any k.
3. If $F = F'$ then all capacities in the network before the change are multiples of k.

Exercise 1: Solution

1. True.

Let S' be a minimal cut. By the Min Cut Max Flow theorem, $F' = C'(S')$.

$$F' = C'(S') = \sum_{e_i \in out(S')} c'(e_i)$$

It is known that the capacity of every edge $e \in E$ is a multiple of k, therefore:

$$F' = \sum_{e_i \in out(S')} c'(e_i) = \sum_{e_i \in out(S')} k \cdot z_i = k \cdot \sum_{e_i \in out(S')} z_i$$

Exercise 1

Prove or disprove:

1. F' is a multiple of k. True
2. **$F \leq F'$ for any k.**
3. If $F = F'$ then all capacities in the network before the change are multiples of k.

Exercise 1: Solution

2. True.

Let S' be a minimal cut. By the Min Cut Max Flow theorem, $F' = C'(S')$.

$$F' = C'(S') = \sum_{e_i \in out(S')} c'(e_i)$$

It is known that the capacity of every edge $e \in E$ was rounded up, therefore:

$$F' = \sum_{e_i \in out(S')} c'(e_i) \geq \sum_{e_i \in out(S')} c(e_i)$$

And S' is a valid $s - t$ cut in the original capacity values

So F' is at least the value of some $s - t$ cut in the original capacity values.

Exercise 1: Solution

2. True.

By the Min Cut Max Flow theorem, F is equal to the value of a minimum $s-t$ cut S in the original capacities.

$$\sum_{e_i \in out(S)} c(e_i) = F$$

Thus

$$F' = \sum_{e_i \in out(S')} c'(e_i) \geq \sum_{e_i \in out(S')} c(e_i) \geq \sum_{e_i \in out(S)} c(e_i) = F$$

Exercise 1

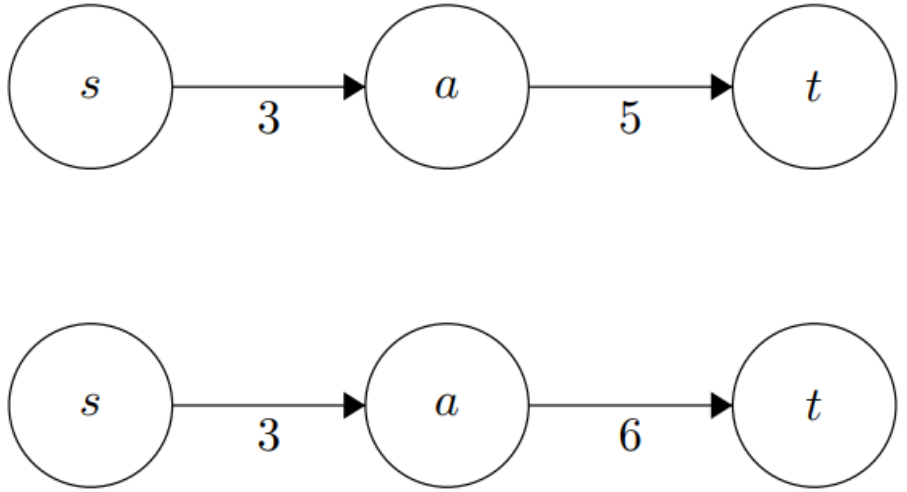
Prove or disprove:

1. F' is a multiple of k. True
2. $F \leq F'$ for any k. True
3. **If $F = F'$ then all capacities in the network before the change are multiples of k.**

Exercise 1: Solution

3. False.

Consider the following case for $k = 3$:

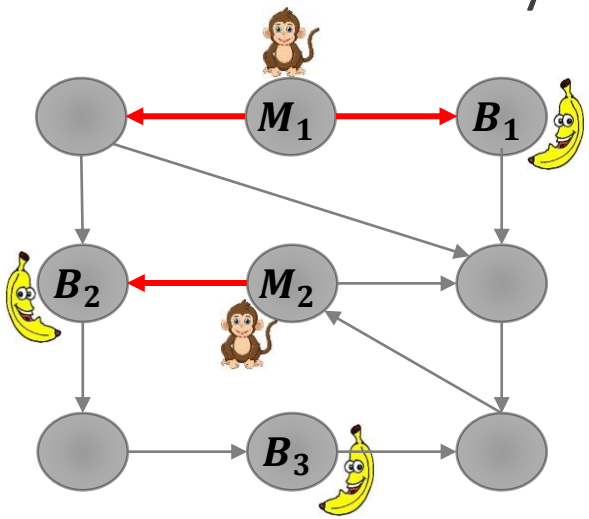


$F = F'$ while the capacities in the network before the change are not multiples of k.

Exercise 2

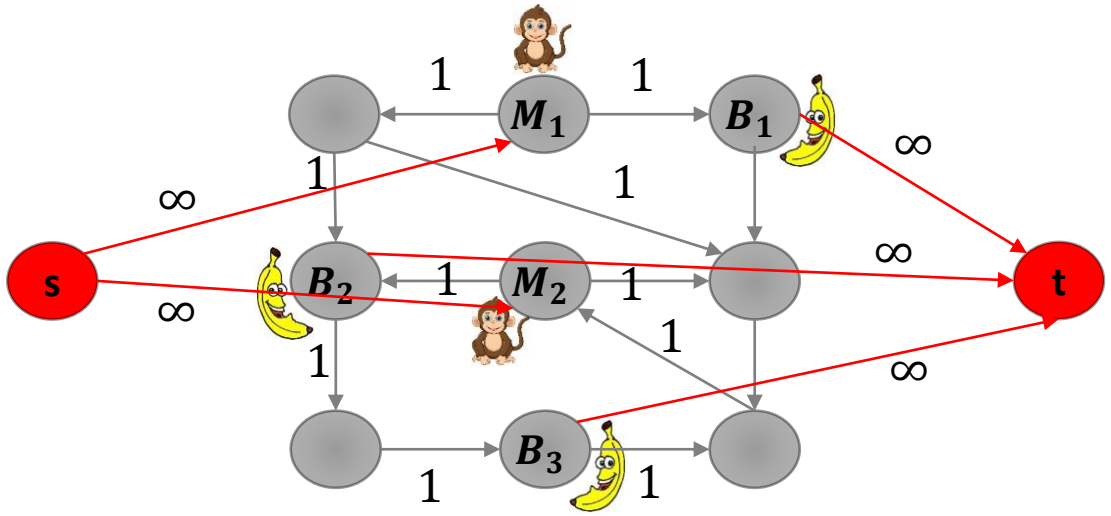
Let $G = (V, E)$ be a directed graph, and let $B, M \subset V$ be two disjoint, non-empty sets of vertices. In every $v \in M$ there is a single monkey and in every $v \in B$ there is a single banana.

1. Suggest a flow-based algorithm to find the **minimal number of edges** we need to remove from the graph such that no monkey can reach a banana.



Exercise 2: Solution

General idea: Build the following network N: add a source and a target to G and connect the source to all monkeys and all the bananas to the target with infinite capacities. Let all other edges in the network have capacity 1.



Exercise 2: Solution

Algorithm:

1. Construct $N = (V', E')$ such that:

$$V' = V \cup \{s, t\}$$

$$E' = E \cup \{(s, v) | v \in M\} \cup \{(v, t) | v \in B\}$$

$$c(e) = 1 \text{ if } e \in E$$

$$c(e) = \infty \text{ if } e \in E' \setminus E$$

2. Run FF (or EK) to find maximum flow from s to t in N.

3. Return the value of the max flow/number of edges in the minimum cut (why are they the same?)

Exercise 2: Solution

Correctness:

We show that the capacity of the min-cut is the minimal number of edges that should be removed in order to separate the monkeys from the bananas.

Let D be the set of edges in the min-cut.

Since $B \cap M = \emptyset$, there is a finite cut in N. This means that the cut D only contains edges from the original graph G, that is $D \subseteq E$.

Exercise 2: Solution

Correctness:

In the graph $G' = (V, E \backslash \mathrm{D})$ there is no path from a monkey to a banana. That is, the vertices of M are separated from the vertices in B. This holds since there is no augmenting path in the residual network.

Moreover, according to the max-flow-min-cut theorem, D is a minimum set of edges having this property.

Because the capacities in the network, $c(e) = 1 \; \forall e \in E$, $c(D) = |D|$.

Exercise 2: Solution

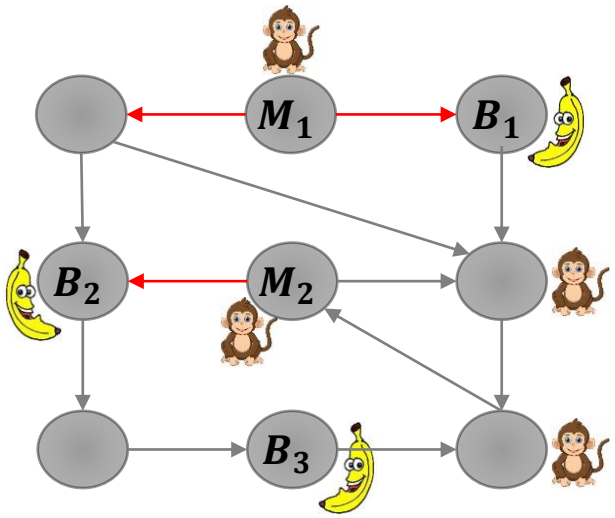
Time complexity:

All capacities are integers thus EK takes $O(\min(m^2 n, m f^*))$.

Since $f^* \leq m$, the algorithm runs in $O(m^2)$.

Exercise 2

2. Suggest an algorithm that detects all vertices in the graph in which a new monkey can be placed such that the answer to question 1 does not change.



Exercise 2

Solution: A monkey can be added to a vertex from which it cannot reach a banana.

Algorithm:

1. Run the algorithm from the answer to question 1.

2. Let R_f be the last residual network.

3. Let T be the vertices in R_f from which there is a path to t (detect T by running BFS from t on R_f with reversed edges).

4. Return $V \setminus (M \cup B \cup T)$.

Exercise 2: Solution

Correctness:

In step 3 we find the min-cut closest to t.

Adding a monkey to a vertex $x \in V \setminus (M \cup B)$ does not change the value of the solution $\Leftrightarrow$

There exists a min-cut C of N such that x and t are in different connected components of $E \setminus C \Leftrightarrow$

t is not reachable from x in R_f.

Exercise 3

A schedule of doctors for the next n days should be detected:

- There are k doctors.
- In the j-th day, r_j doctors should be assigned.
- Each doctor i has a list $L_i \subseteq \{1, \dots, n\}$ of days she can work on.
- Each doctor can be assigned to work on at most $Z \leq n$ days.

Exercise 3

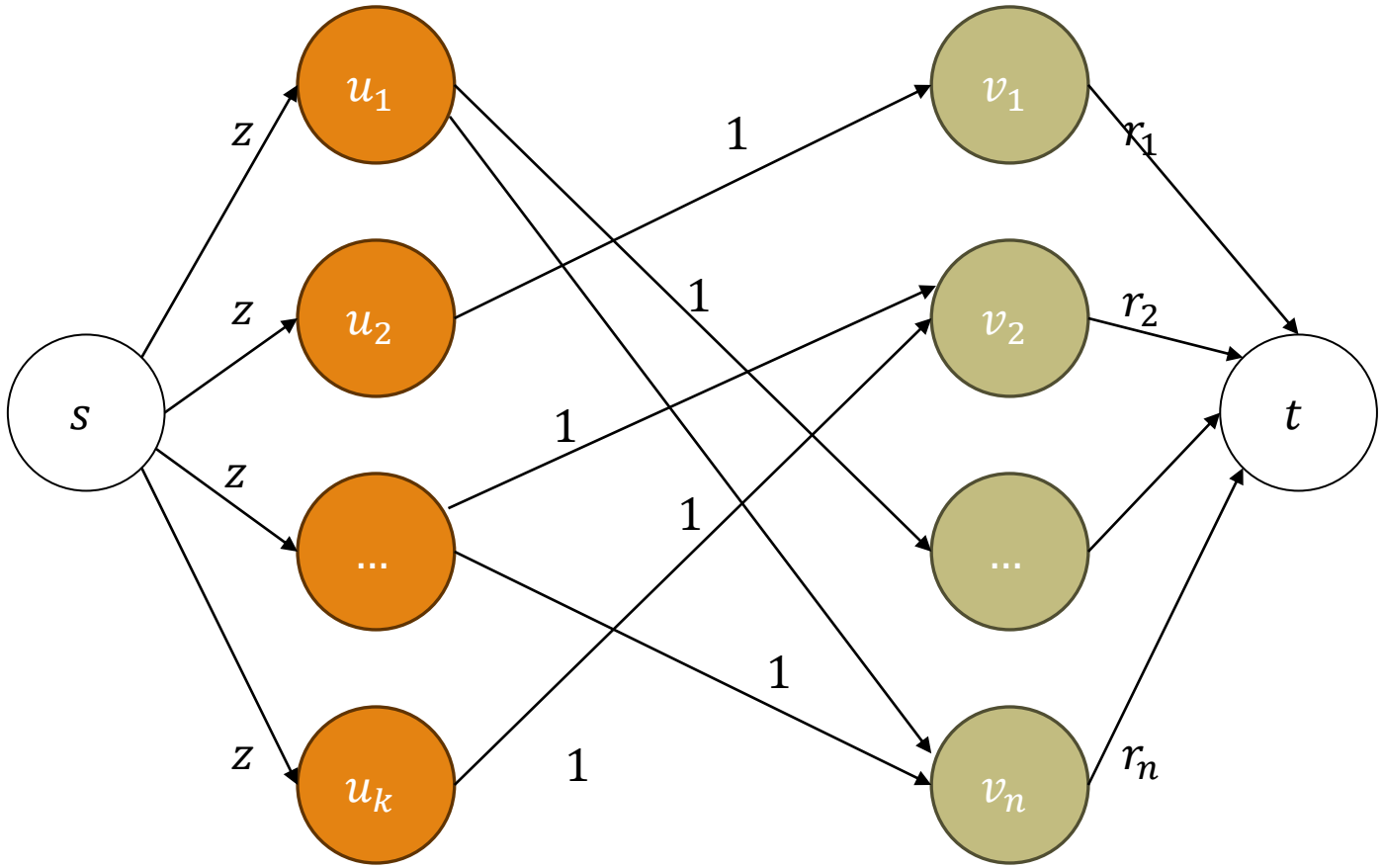
Suggest an efficient algorithm that finds, *if possible*, the set of days in which every doctor is assigned. Formally, return for each $1 \leq i \leq k$, a set $D_i \subseteq \{1, \ldots, n\}$ such that:

- $\forall\ 1 \leq i \leq k, D_i \subseteq L_i$.
- $\forall\ 1 \leq i \leq k, |D_i| \leq Z$.
- $\forall\ 1 \leq j \leq n$, exactly r_j doctors are assigned.



Exercise 3: Solution

Exercise 3: Solution



k doctors

n days

Exercise 3: Solution

1. Construct the following flow network G:

$$V = \{s, t\} \cup \{u_i\}_{i=1}^{k} \cup \{v_j\}_{j=1}^{n}$$

$$E = \{(s, u_i)\ \forall i\} \cup \{(u_i, v_j) : j \in L_i\} \cup \{(v_j, t) \forall j\}$$

Where the capacities are as follows:

- $\forall i, (s, u_i) = \mathrm{z}$
- $\forall i\ , j$, such that j $\in L_i, (u_i, v_j) = 1$
- $\forall j, (v_j, t) = \mathrm{r_j}$

Exercise 3: Solution

2. Find a maximum flow in G.

3. If $f(v_j, t) = r_j$ for every $1 \leq j \leq n$ then there is a valid solution. Return for every doctor

$$D_i = \{j : f(u_i, v_j) = 1\}$$

Otherwise, return “assignment is not possible”.

Exercise 3: Solution

Correctness:

Main claim: f is a maximum flow in G with $f(v_j, t) = r_j$ for each j **if and only if** $\{D_i\}$ is a valid schedule.

Claim 1: If f is a valid flow with $f(v_j, t) = r_j$ for each j then $\{D_i\}$ is a valid schedule.

Claim 2: If $\{D_i\}$ is a valid schedule, then the cut $(V \setminus \{t\}, \{t\})$ is saturated in any maximal flow.

Exercise 3: Solution

Correctness:

Claim 1: If f is a valid flow with $f(v_j, t) = r_j$ for each j then $\{D_i\}$ is a valid schedule.

Proof: Since the flow is in integers and the edge (u_i, v_j) maintains that $(u_i, v_j) \in E$ iff $j \in L_i$, we get for any valid flow: $D_i \subseteq L_i$.

Moreover, since $c(s, u_i) = Z$, a doctor can be assigned in at most Z days. According to the edge rule, we assigned $f(v_j, t)$ doctors to the j-th day. Since $f(v_j, t) = r_j$, the schedule is valid.

Exercise 3: Solution

Correctness:

Claim 2: If $\{D_i\}$ is a valid schedule, then the cut $(V \setminus \{t\}, \{t\})$ is saturated in any maximal flow.

Proof: Let f be the following flow:

$f(s, u_i) = |D_i|, f(u_i, v_j) = 1 : \text{j} \in D_i, f(v_j, t) = r_j$

f is a valid flow since $\{D_i\}$ is a valid schedule.
Since $F = C_{out}(V \setminus \{t\})$, the cut is saturated and f is indeed a maximum flow.

Thus, $V \setminus \{t\}$ is a minimum cut, and is saturated in any maximal flow.

Exercise 3: Solution

Time complexity:

$$|V| = O(n + k), |E| = O(nk), f^* \leq nk$$
$$\Rightarrow f^* = O(|\mathrm{E}|) \leq |E| \cdot |V|.$$

Running FF takes $O(\min(|E|^2|V|, |E|f^*)) = O(n^2k^2)$ time.

Exercise 3, cont.

Assume that a valid schedule $\{D_i\}$ is detected (and **is given to you**).

It turns out that r_1 should be increased by one, and r_2 should decrease by one.

$$r_1' = r_1 + 1, r_2' = r_2 - 1$$

Suggest an **efficient** algorithm to calculate a new valid assignment (if exists).

Exercise 3, cont. – solution 1

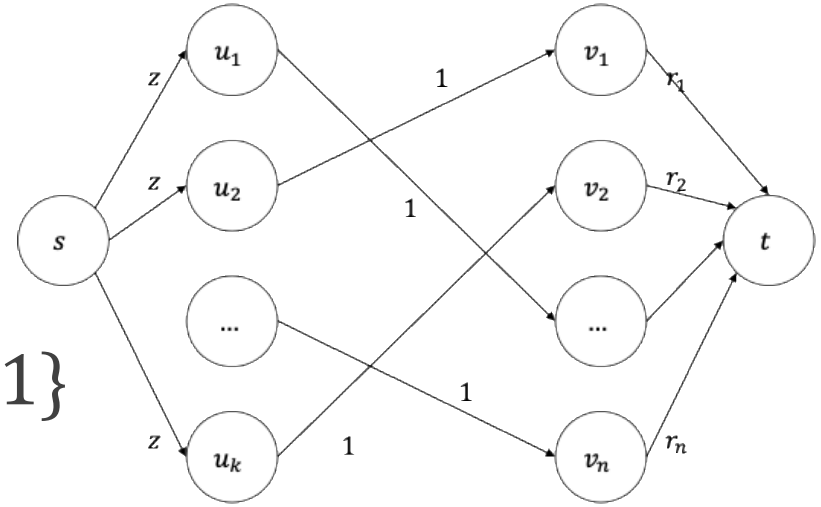
1. Construct the flow network G again (or update r_1 and r_2)
2. Find a maximum flow in G.
3. If $f(v_j, t) = r_j$ for every $1 \leq j \leq n$ then

   there is a valid solution.

   Return for every doctor $D_i = \{j : f(u_i, v_j) = 1\}$

Otherwise, return “assignment is not possible”.

<u>Complexity</u>: $O(n^2k^2)$



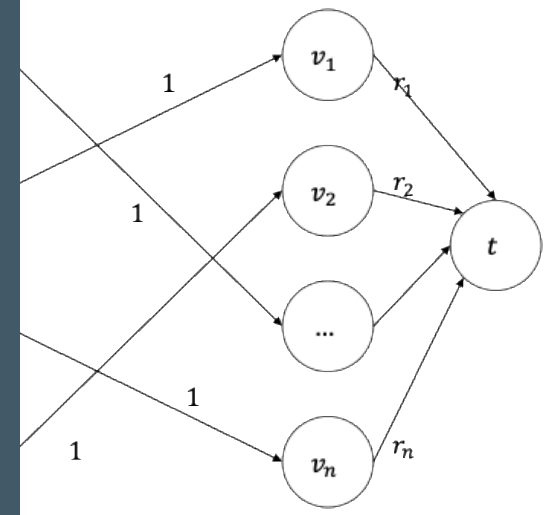
Exercise 3, cont. – solution 1

1. Construct the flow network G again (or update r_1 and r_2)
2. Find a m
3. If $f(v_j, t$
   there is a
   Return fo

Otherwise,

Complexity: $O(n^2k^2)$





Exercise 3, cont. – solution 2

Consider the last residual network.

1. Run Breadth-First Search (BFS) starting from node v_2.

2. If v_1 is reached, then a modification of the schedule is possible:

Augment 1 unit of flow along the path from v_2 to v_1.

Note: The path from v_2 to v_1 can use any node in the residual graph (doctors, other days, or even the source s), as long as the path ends at v_1.

Time complexity: $O(|E|) = O(nk)$.

3. If v_1 is not reached, then a modification of the schedule is not possible (proof left for the reader.)

Exercise 3, cont. – solution 2

Correctness:

Claim: If v_1 is reached, the modified schedule is valid.

Proof: The modification takes the original max flow f and creates a new flow function f', by augmenting 1 unit of flow along the path from v_2 to v_1.

The new flow f' is a legal flow in the network with r_1, r_2 updated to r_1', r_2' respectively (respects the edge capacities and vertex flow conservation).

The cut $(V \setminus \{t\}, \{t\})$ is a saturated min cut under the new flow f', making it a max flow. By the original claims (1) and (2), the modified schedule is legal.